

Design a URL Shortener - Back of the Envelope Estimation

1. Write Traffic

None

100 million URLs / day

- Write per second

None

$100M / (24 \times 3600) \approx 1160 \text{ writes/sec}$

2. Read Traffic

- Assume read : write = 10 : 1

None

$\text{Read/sec} = 1160 \times 10 = 11,600 \text{ reads/sec}$

3. Data Growth

None

$100M \times 365 \times 10 \text{ years} = 365 \text{ billion URLs}$

4. Storage Assumption

- Average URL length:

None

100 bytes

5. Total Storage

None

$365B \times 100 \text{ bytes} = 36.5 \text{ TB per year}$

None

$\text{For } 10 \text{ years} = 365 \text{ TB}$

6. Key Takeaway

System must handle:

- ~1160 writes/sec
- ~11,600 reads/sec
- ~365 billion records
- ~365 TB storage over 10 years